Verification and Approximation Study for *Generalized Cluster Dynamics*

RadCluster_2_1

11 September 2026

Abstract

This report presents a verification and approximation study of the bin-moment reduction used in the generalized cluster-dynamics model of ferritic–martensitic steels, in response to the §6.2 / Figure 8 critique. The reduction is examined along each of its parameters — the extent of the discrete core, the number of logarithmic bins, the closure order and the integration tolerance — and then compared with the fully discrete solution of the same equations on a domain small enough for that solution to be obtained. Six observable quantities are reported for every configuration, against EUROFER97 neutron measurements at 300 °C to 350 °C where the comparison is meaningful. A second group of calculations separates the two parameters the first varied together, system size and bin count, and shows that the one quantity failing to converge in bin count, the $\frac{1}{2}\langle 111\rangle$ mean loop diameter, converges in the extent of the discrete core. Three defects found in the course of the work — a physics channel absent from one solver path, a dose ladder that scored at the wrong dose, and a corrupted moment reaching the interstitial inventory — are reported with the results they affect.

Contents

1 Convergence of the Bin-Moment Approximation **2**
1.1 Computational configuration and reference data . . . 2
1.2 Refinement of the discrete core . . . 3
1.3 Bin count, closure order and integration tolerance . . . 3
1.4 Treatment of helium . . . 4
1.5 Verification against the exact discrete solution . . . 4
1.6 Extension of the reduced grids to the measured dose range . . . 5
1.7 Conservation diagnostics and incomplete configurations . . . 6

2 Effects of System Size **6**
2.1 Dose dependence along the verification ladder . . . 7
2.2 Separating system size from bin count . . . 10
2.3 Cost of the exact reference solution . . . 10
2.4 Bin count at a fixed discrete core . . . 10
2.5 The two-dimensional sweep . . . 12
2.6 Origin of the residual mean-size error . . . 12

1 Convergence of the Bin-Moment Approximation

The bin-moment reduction replaces the discrete master equation over a domain of 10^4–10^5 cluster sizes with a hybrid representation: sizes up to i_{discrete} are carried individually, and the remainder is partitioned into I_{bin} logarithmically spaced bins carrying only the leading moments, the intra-bin distribution being reconstructed from a shape function of order P. Vacancies are treated identically, with v_{discrete} and V_{bin}.

No single calculation establishes that such a reduction may replace the full system. This section proceeds through four groups, each varying one feature of a common reference configuration. T1 refines the reduction's own resolution parameter, i_{discrete}: if the reduction is converged, the observables must cease to move as the discrete core is enlarged. T2 varies the remaining numerical parameters one at a time — I_{bin}, the closure order P and the relative tolerance. T3 tests a reduction in the physics rather than the numerics: the quasi-steady-state treatment of helium against dynamic integration. T4 is the verification proper, comparing the reduced solution with the exact discrete solution on a domain small enough for the latter to be obtained; T4b carries the same grids to the dose at which the measurements lie.

T1 to T3 establish internal consistency; only T4 establishes accuracy, and at a single domain and dose. That restriction motivates Section 2, which follows the T4 ladder in dose and then separates the two axes T4 varied together.

1.1 Computational configuration and reference data

All calculations share the EUROFER97 host declaration, the fission cascade spectrum, $G = 1 \times 10^{-7}$ dpa/s, 330 °C, `full_system` solver mode, GMRES with the Woodbury preconditioner, `atol` $= 10^{-20}$, and $i_{\mathrm{mobile}} = v_{\mathrm{mobile}} = 5$; what differs is listed in Table 1. The labels T1–T4 and T4b correspond to the run identifiers `T1_B2`, `T2_P1` and so forth in the archived output, while "Table N" denotes a numbered table of this report.

Table 1: Configuration of each group. N_{eq} is the main block; the ⟨100⟩ loop block is carried separately and adds a further I equations in discrete mode, so run D integrates 12 006 rather than 8 006.

Block	Equations	I	V	i_{discrete}	I_{bin}	Dose (dpa)	Scored at
T1 (B2–B4)	bin_moment	80 000	20 000	1600–100	11–18	40	15.72 dpa
T2	bin_moment	80 000	20 000	100	10–18	40	15.72 dpa
T3	bin_moment	80 000	20 000	100	18	40	15.72 dpa
T4 (D)	**discrete**	4 000	4 000	—	—	2	2.000 dpa
T4 (C1–C4)	bin_moment	4 000	4 000	400–5	6–18	2	2.000 dpa

The experimental reference is the set of EUROFER97 neutron measurements at 300 °C to 350 °C compiled in Table 2. Their dose coverage matters: the loop data begin at 14.6 dpa, which is why T1–T3 are scored at 15.72 dpa and why T4, integrated only to 2 dpa, cannot be compared with them.

One measurement requires comment. A single ⟨100⟩ row reports a mean loop diameter of 48 nm at 17.4 dpa and 350 °C, against 2.8 nm to 8.0 nm for the other nine rows; its density, 3.7×10^{21} m^{-3}, is unremarkable. The full range is retained in the tables, but this diameter is excluded from the figures of Section 2, where it would set the ordinate on its own.

A defect in the scoring procedure invalidated a first round of results. The output grid held no point at the requested scoring dose, so a nominal 15.72 dpa was scored at 11.51 dpa, 26.8 % low,

Table 2: EUROFER97, neutron irradiation, 300 °C to 350 °C. Measured ranges over all sources in `docs/Database`: loops from [1, 2, 3, 4, 6], cavities from [1, 4, 5, 6, 7].

Population	n	Density (10^{21} m^{-3})	Diameter (nm)
$\frac{1}{2}\langle 111\rangle$ loops	13	0.30 – 34.0	3.3 – 35.0
$\langle 100\rangle$ loops	10	2.24 – 12.0	2.8 – 48.0
Cavities	6–8	0.32 – 6.30	1.6 – 16.0
Dose coverage: loops 14.6 dpa to 32 dpa, cavities 2.7 dpa to 32 dpa			

while the run reported having reached the requested value. Every affected row has been recomputed on a grid containing the scoring point. This is the same failure mode as the Figure 8 defect the study was commissioned to address, found within the study's own tooling.

1.2 Refinement of the discrete core

The first group refines i_{discrete} at the production domain, $I = 80\,000$ and $V = 20\,000$, with the bin count adjusted to hold the bin ratio approximately constant as the core contracts. The three rungs that reached the comparison dose are given in Table 3.

Table 3: Refinement of the discrete core at $I = 80\,000$, $V = 20\,000$, scored at 15.72 dpa. B4 is the tracked reference. Rungs that did not reach the comparison dose are omitted; see Section 1.7.

Rung	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)	δ_{FP}
B2 (i_d 1600)	2048	3.449	4.892	4.156	6.001	2.233	2.232	0.069
B3 (i_d 400)	562	3.512	4.509	4.090	5.868	2.237	2.234	0.068
B4 = production	**189**	**3.498**	**4.637**	**4.715**	**5.939**	**3.609**	**2.890**	**0.060**
Experiment		≥0.30	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16	—

Across B2, B3 and B4 the observables move by a few percent while N_{eq} falls by an order of magnitude: d_{100} spans 5.868 nm to 6.001 nm and N_{111} from 3.449 to 3.512 — the behaviour expected of a converged reduction under refinement of its own resolution parameter, and the principal claim the group was designed to support.

Two qualifications attach. First, N_{cav} and d_{cav} at B4, 3.609 and 2.890, lie above their B2 and B3 values of 2.233 and 2.232 by more than the loop observables move, so the cavity population is the least converged. Second, the most highly resolved rung planned, $i_d = 6400$, did not reach the comparison dose, so convergence is demonstrated over a tenfold span in N_{eq} rather than the fortyfold planned.

1.3 Bin count, closure order and integration tolerance

The second group varies one parameter at a time from the B4 reference, holding $i_{\text{discrete}} = 100$ (Table 4).

The two tolerance columns, `rtol` $= 1 \times 10^{-4}$ and 1×10^{-6}, agree in every digit reported: two orders of magnitude in integration tolerance move no observable, so the tolerance is not a free parameter concealed within the results, one of the specific concerns raised in the review. Reducing the bin count from 18 to 10 likewise displaces the observables by a few percent at most.

Table 4: One parameter varied at a time from the B4 reference configuration, scored at 15.72 dpa.

Column	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)	δ_{FP}
B4 (reference)	189	3.498	4.637	4.715	5.939	3.609	2.890	0.060
$P = 1$ constant	150	2.566	**15.580**	5.335	**45.676**	11.361	2.751	0.158
I_{bin} 10	171	3.543	4.282	4.021	5.861	2.900	2.340	0.070
`rtol` 1×10^{-4}	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
`rtol` 1×10^{-6}	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
Experiment		$\geq$0.30	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16	—

Against that flat background the closure order is conspicuous. The piecewise-constant closure, $P = 1$, returns $d_{100} = 45.676$ nm against 5.939 nm and $d_{111} = 15.580$ nm against 4.637 nm — errors of roughly a factor of eight — and degrades δ_{FP} to 0.158, so the existing §6.2 claim is supported by a wide margin. The lognormal closure, $P = 3$, carries no row, having advanced only to 5.4×10^{-4} dpa of the required 40 at 2.9 h per output step, itself the result the design anticipated.

1.4 Treatment of helium

The third group compares the quasi-steady-state reduction of the helium sub-system with its dynamic integration at otherwise identical settings (Table 5) — model against model, not against measurement.

Table 5: Treatment of helium, scored at 15.72 dpa; model against model. The fusion arms are omitted: one did not reach the comparison dose, the other is blocked on the §3.1 `he_model` unwelding, and neither has a corresponding EUROFER97 dataset.

Column	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)	δ_{FP}
fission QSS	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
fission dynamic	190	3.533	4.354	4.153	5.919	3.253	2.434	0.068

The two treatments agree in every digit: 3.533, 4.354, 4.153, 5.919, 3.253 and 2.434. The quasi-steady-state reduction may therefore be used under fission without consequence for any of the six observables, there being almost nothing to integrate in time — the reference calculation holds $C_{\text{He,tot}} = 1.18 \times 10^{13}\,\text{m}^{-3}$.

This is a null and should be read as one: taken alone it is equally consistent with a helium treatment that does not matter and with a comparison never sensitive to it. Distinguishing the two requires a fusion arm, and neither is reported here — one failed to reach its comparison dose, the other is blocked on the §3.1 `he_model` unwelding. The safety of the reduction is established at fission helium levels only.

1.5 Verification against the exact discrete solution

The preceding groups compare reduced solutions with one another; this one compares them with the exact solution of the same equations, on a reduced domain $I = V = 4\,000$ chosen so the discrete system remains tractable. Deviations in Table 6 are relative to run D, not to experiment.

An important restriction applies. Production runs with `LOOP_COAL` = 1, enabling the $\frac{1}{2}\langle 111\rangle$ loop–loop coarsening channel, which was originally implemented only in the bin-moment path; the discrete path accepted the flag, recorded it in provenance, and ignored it. The discrete arm was therefore never the exact solution of the same equations, and the first version of this comparison, appearing to show a closure error of 90 to 114 % in d_{111}, was measuring the absent channel. The channel has since been implemented in the discrete path and the two verified to agree to 7×10^{-8} (`codes/Python_Testing/check_bm_vs_discrete.py`), but an exact arm in the production configuration proved unaffordable: the term is $O(N_C^2)$ in the populated sessile sizes and, even gated, consumed 2.8 h on one output step at 0.014 dpa of a required 2. Both arms are therefore run with the channel disabled, and the closure error measured here is not demonstrably that of the production configuration. Without the channel the $\frac{1}{2}\langle 111\rangle$ mean size also pins at the mobility cutoff and its density exceeds the measurements 20- to 60-fold, so nothing in Table 6 may be compared with experiment — though this does not invalidate a verification, which asks only whether the reduction reproduces the exact solution of the same equations.

Table 6: Verification against the exact discrete solution at $I = V = 4\,000$, with `LOOP_COAL = 0` in both arms, scored at 2.0000 dpa. Percentages are deviations from run D. N_{eq} is the main block, which is what the solver reports; *total* adds the appended $\langle 100\rangle$ loop block that CVODE also integrates. The figures of Section 2 are labelled with the total.

Rung	N_{eq}	total	N_{111} (10^{21})	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)
D (exact)	**8006**	**12006**	**12.568**	**1.065**	**5.741**	**2.948**	**2.544**
C1 (1/10)	830	1242	12.335 (−1.9%)	1.044 (−2.0%)	5.690 (−0.9%)	2.555 (−13.3%)	2.273 (−10.6%)
C2 (1/40)	244	364	12.144 (−3.4%)	1.044 (−2.0%)	5.681 (−1.0%)	2.574 (−12.7%)	2.279 (−10.4%)
C3 (1/160)	108	161	7.027 (−44.1%)	1.018 (−4.4%)	5.349 (−6.8%)	3.219 (+9.2%)	2.451 (−3.6%)
C4 (1/800)	86	129	6.142 (−51.1%)	0.953 (−10.5%)	5.678 (−1.1%)	6.575 (+123.0%)	2.660 (+4.6%)
C4, $P = 1$	51	94	17.498 (+39.2%)	0.059 (−94.5%)	10.755 (+87.3%)	20.929 (+610%)	2.632 (+3.5%)

Against the fully discrete solution the linear closure holds d_{100} to within 1.1 % at every rung from 830 equations down to 86 — a hundredfold reduction in cost for a one-percent error — and N_{100} to 10.5 % at the coarsest rung, 2 % at the finest.

The exceptions are informative. N_{111} degrades sharply below C2, by 44 % at C3 and 51 % at C4: the $\frac{1}{2}\langle 111\rangle$ population is peaked at small sizes, and once i_{discrete} falls to 25 and then 5 that peak lies inside the binned region, where the closure represents it rather than resolving it. N_{cav} is worse still at C4, 123 %. The coarsest rung, which is the production resolution, buys its speed at real cost in the two *density* observables, while the *sizes* remain accurate.

Convergence along this ladder is not monotonic: C1 and C2 are nearly identical, C3 is worse than C2 in both N_{111} and d_{100}, and C4 recovers in d_{100} while degrading further in N_{cav}. A ladder that does not improve monotonically is not evidence of a converged closure, and is presented as an observation rather than as one. The likeliest explanation, tested in Section 2, is that i_{discrete} and the bin partition interact: the rungs hold the bin ratio r fixed, but the discrete/binned boundary moves relative to the peak of the distribution. The piecewise-constant closure fails against the exact solution as decisively as against B4, by 87 % in d_{100} and −94.5 % in N_{100}.

1.6 Extension of the reduced grids to the measured dose range

The C2 to C4 grids were integrated to the production horizon, so they may be compared with experiment at 15.72 dpa — which the preceding group cannot, being scored at 2 dpa where no loop data exist. Run D has no counterpart: the discrete arm could not be carried beyond 0.014 dpa, so above 2 dpa these grids have no exact reference. Table 7 therefore exhibits dose dependence and the approach to the measured band, not verified accuracy.

Table 7: The no-coarsening grids integrated to 40 dpa and scored at 15.72 dpa against the EUROFER97 measured ranges. This is not a verification: no exact solution exists at this dose.

Rung	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)
C2 (1/40)	244	13.092	1.722	6.916	5.045	1.951	2.135
C3 (1/160)	108	6.770	1.487	6.392	4.770	2.235	2.230
C4 (1/800)	86	5.276	1.476	5.950	4.748	3.431	2.484
Experiment	—	0.30–34.0	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16

All three grids fall within the measured ⟨100⟩ density and diameter ranges and the cavity diameter range. N_{111} lies above the measured band and d_{111} below it — the signature of the disabled coarsening channel, and an expected consequence of that variant rather than an independent finding.

1.7 Conservation diagnostics and incomplete configurations

Two conservation defects run through the results above and are reported rather than corrected. The Frenkel-pair diagnostic δ_{FP} lies between 0.05 and 0.07 in all columns, and reaches 0.158 in the piecewise-constant columns, against the model's own gate of 10^{-2}. Closure error and conservation error are distinct quantities and should not be conflated.

The second defect inflates $C_{\mathrm{SIA,tot}}$ by a factor of 17 to 25 in fine-bin runs, $I_{\mathrm{bin}} \geq 16$: when the uppermost bin empties its zeroth moment falls to the C_{floor} level while its first moment does not, leaving $\mu_1/\mu_0 \sim 10^8$ in a bin spanning sizes 946 to 1001. `_floor_bin_moments` clamps μ_1/μ_0 back inside the bin on the mean-size path, but `SIA_content_from_mu1` sums the raw first moment, so the corruption reaches $C_{\mathrm{SIA,tot}}$ unguarded. The six observables, δ_{FP} and the swelling are unaffected; what fails is the swelling identity $S = S_I + \Delta J^d$, in which $C_{\mathrm{SIA,tot}}$ is S_I.

Five configurations did not reach their comparison dose and are absent from the tables: the $i_d = 6400$ rung of T1 (0.079 dpa of 40 after 18.9 h); the $P = 3$ lognormal closure of T2 (5.4×10^{-4} dpa) and its $I_{\mathrm{bin}} = 40$ column (5.70 dpa); the fusion arm of T3 with dynamic helium (0.052 dpa after 13.9 h); and the C0 rung of T4, which grows stiffer the more of the frozen spectrum is resolved discretely. A sixth, the fusion Case 1/2 arm of T3, was never runnable. The reporting tool refuses to emit a metric for any of them, so the rule is mechanical rather than a matter of discipline.

The null tests of preconditioner and thread count were not performed: they take wall-clock time as the dependent variable and so require an otherwise idle machine. Existing evidence stands in their place. In `compare_linsol_results.json`, the one pair in which both arms ran to completion (bin-moment, $N_{\mathrm{eq}} = 190$, 0.01 dpa) agrees to nine significant figures in $\langle n_i \rangle$ — 24.563 925 879 7 with Woodbury against 24.563 925 898 6 with Jacobi — seven in $\langle m_v \rangle$ and eight in the swelling, only the wall-clock time differing, by a factor of 1.29.

2 Effects of System Size

The preceding section establishes accuracy at a single domain and dose. This one examines how that accuracy depends on domain size: first by following the verification ladder of Section 1.5 in dose, then through a second group of calculations, designated M, separating domain size from bin count.

2.1 Dose dependence along the verification ladder

Figures 1 to 5 show the dose dependence of each observable across the ladder, ordered D, C4, C3, C2, C1, with the measurements and the band spanning them overlaid. The $\frac{1}{2}\langle 111\rangle$ mean diameter is excluded: with the coarsening channel disabled it is pinned at the mobility cutoff and carries no information about the reduction.

The trajectories terminate at two different doses, deliberately: C2, C3 and C4 were integrated to 40 dpa to reach the measurements, while D and C1 stop at 2 dpa, the discrete arm having proved impossible to carry further at any affordable cost. Below 2 dpa the coarse grids may therefore be judged against the exact solution, and above it they may not: where the extended trajectories cross the experimental band there is no exact reference, so agreement there speaks to the model rather than to the reduction.

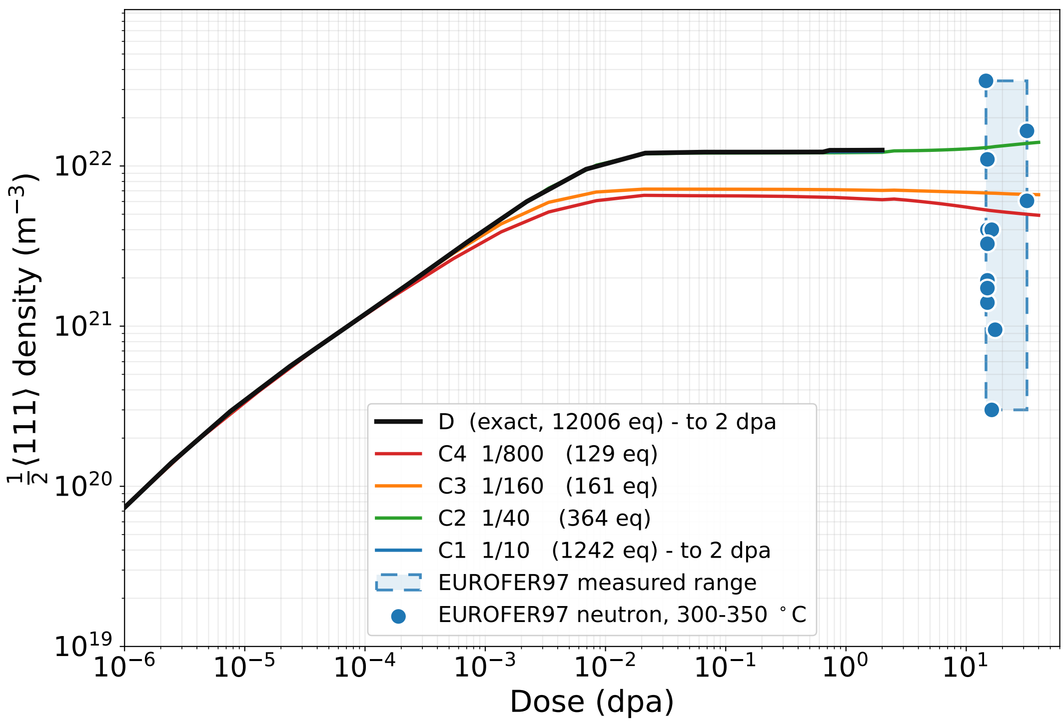

Figure 1: $\frac{1}{2}\langle 111\rangle$ loop number density. The exact solution separates from the coarse grids from 0.1 dpa onwards; C3 and C4 lose approximately half of the population. Measurements from [1, 2, 3, 4, 6].

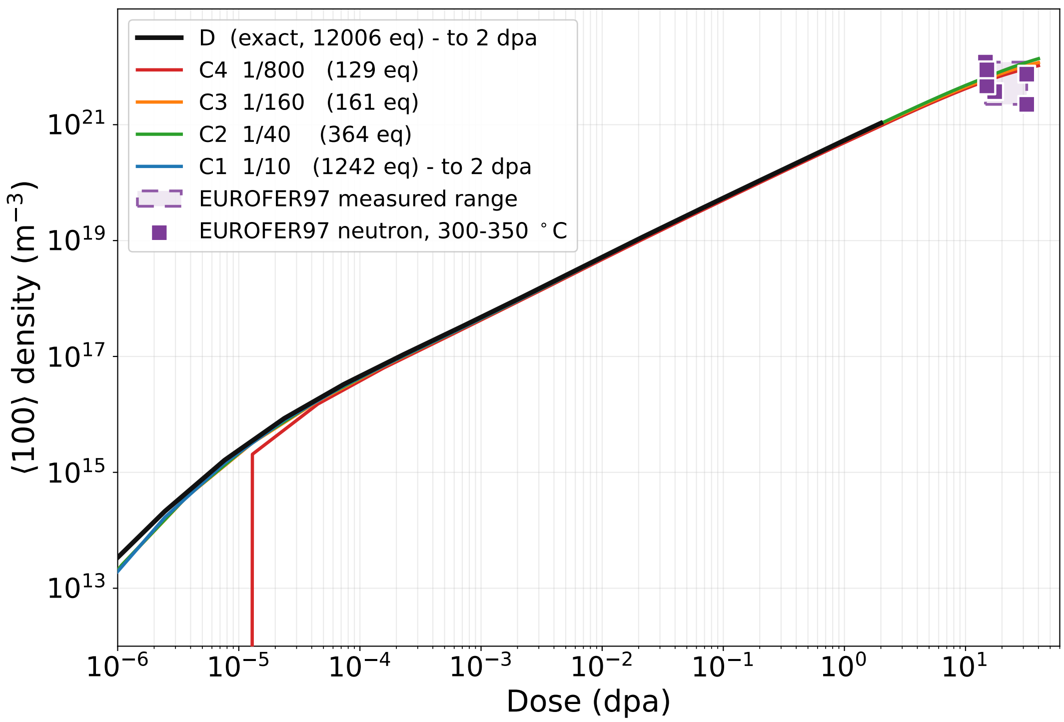

Figure 2: ⟨100⟩ loop number density. All grids track the exact solution closely; this is the observable the reduction represents best. Measurements from [1, 2, 4, 6].

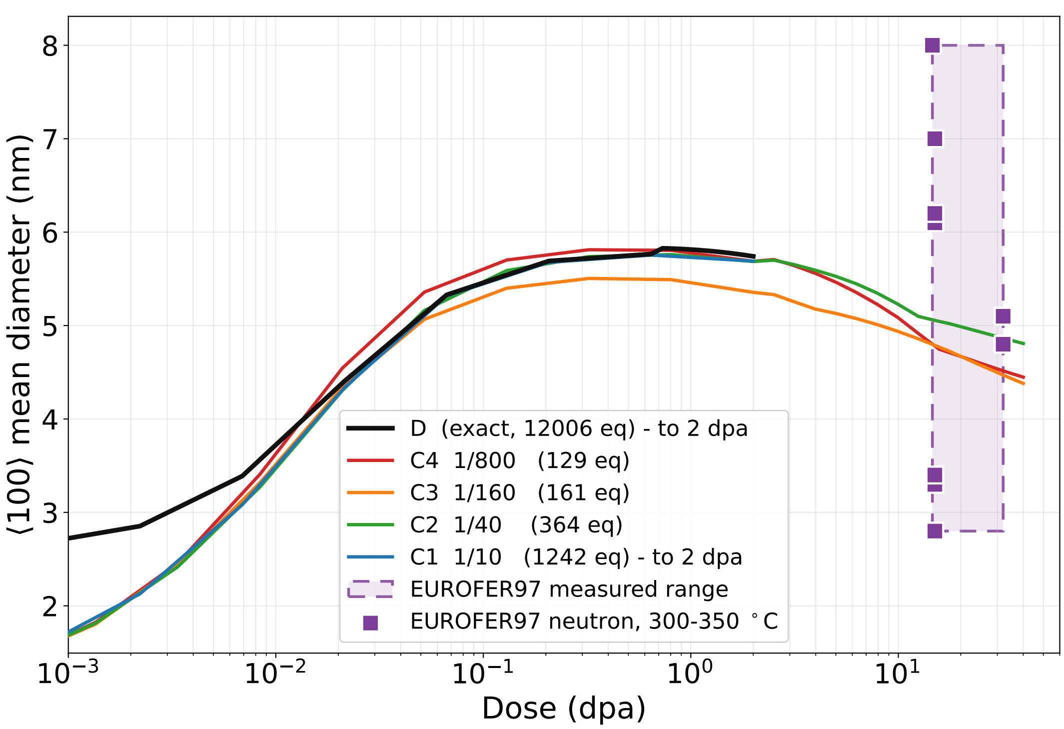

Figure 3: ⟨100⟩ mean diameter, held to about one percent across the whole ladder. Measurements from [1, 2, 4, 6]; the single 48 nm value, from [4], is excluded as described in Section 1.

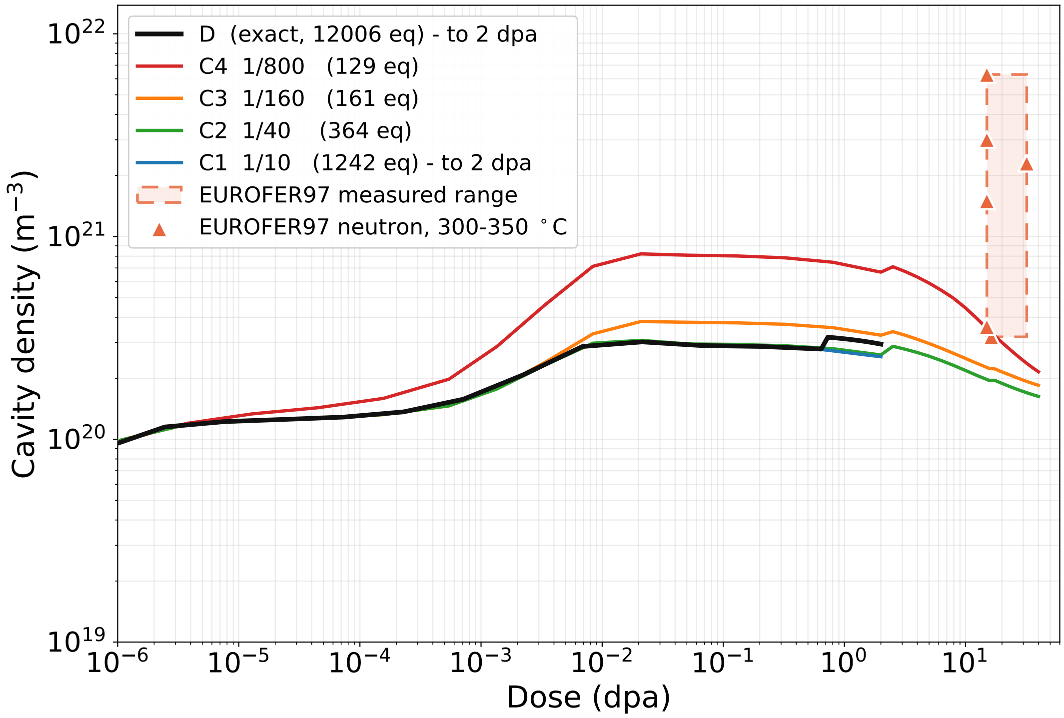

Figure 4: Cavity number density, the observable most sensitive to system size, with C4 lying more than a factor of two above the exact solution. Measurements from [1, 4, 5, 6].

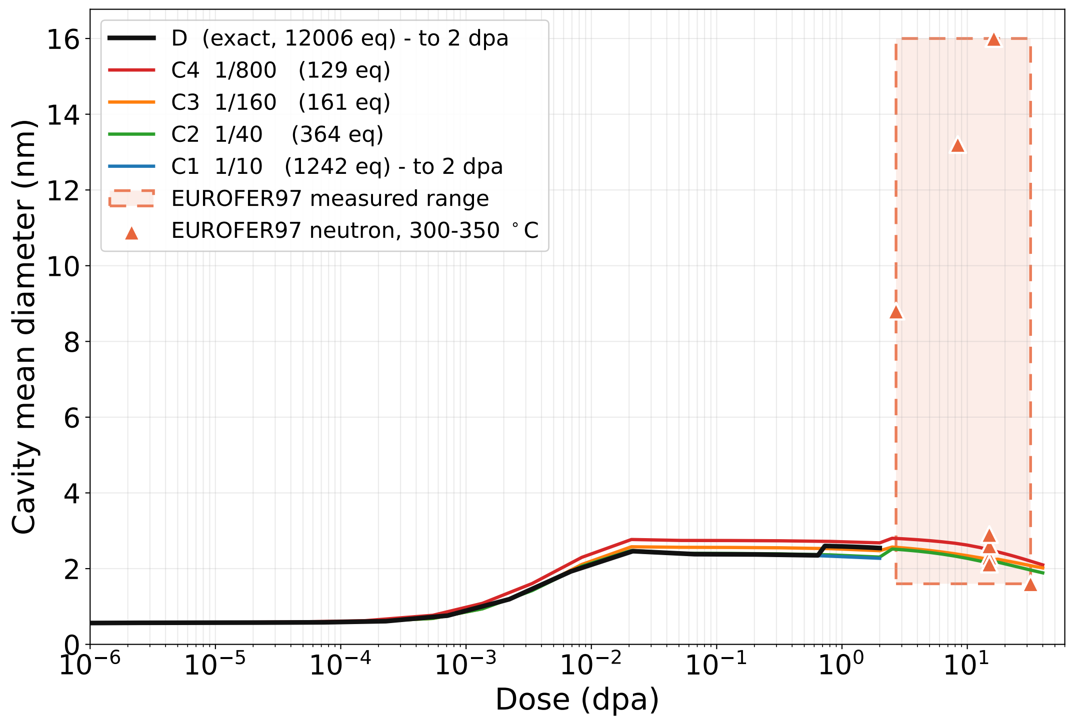

Figure 5: Cavity mean diameter. These measurements extend down to 2.7 dpa, so this is the one panel in which the band and the calculated trajectories nearly meet. Measurements from [1, 4, 5, 6, 7].

2.2 Separating system size from bin count

The T4 rungs moved i_{discrete} and I_{bin} together, so a deviation there cannot be attributed to either. The M calculations separate them: M1 varies the domain with no closure at all, M2R the bin count at fixed discrete core, and M2D both independently, so every cell has a neighbour differing in one axis. All run at production mobility ($i_{\text{mobile}} = v_{\text{mobile}} = 5$), fission cascade, with `LOOP_COAL` and `prec_bw` at their defaults in both arms, on one shared 45-point logarithmic grid to 20 dpa.

The reference domain is small: with $V = 1000$ a cavity is capped at about 2.8 nm, and the exact arm reports $d_{\text{cav}} = 1.74$ nm at 20 dpa, 62 % of that ceiling. Both arms sit on the same truncated domain, so the truncation cancels from the closure error measured, but nothing below may be compared with experiment.

2.3 Cost of the exact reference solution

Table 8 records the cost of the fully discrete arm against domain size.

Table 8: Cost of the fully discrete arm against domain size. The emphasised row is the only one that completes, and is the reference for what follows.

Run	$I = V$	N_{eq}	Dose reached	Wall
M1_D1000	1000	2006	20 / 20	5.2 min
M1_D2000	2000	4006	0.03839 / 20	61.1 min
M1_D4000	4000	—	*not run — see text*	

The increase is abrupt. Doubling the domain takes the exact arm from completing in five minutes to reaching 0.2 % of the same dose in an hour; at $I = V = 10\,000$ it diverges outright, the per-step cost growing roughly fourfold per output step and reaching 2643 s on step 28 of 45, at 104 % CPU and therefore entirely within the serial region. That region is the $O(N_C^2)$ pair sum of `loop_coal`: the participation gate admits sessile sizes above $2\,C_{\text{floor}}$, a population ten times larger at $I = 10\,000$ than at $I = 1000$, so the pair count is a hundred times greater. This is the same failure that defeated the exact arm of Section 1.5 three times. Domains of 4000 and above were not attempted.

2.4 Bin count at a fixed discrete core

Table 9 gives the deviation from the exact arm as the bin count varies at the reference domain, $i_{\text{discrete}} = v_{\text{discrete}} = 50$ throughout.

Five of the six observables converge, as Figure 6 shows: by $I_{\text{bin}} = 12$, where $N_{\text{eq}} = 158$ against 2006 for the exact arm, N_{100}, d_{100}, N_{cav} and d_{cav} all lie within 0.3 % of the exact solution, and N_{111} plateaus at a residual 1.0 %.

The $\frac{1}{2}\langle 111\rangle$ mean diameter does not. It remains between −5 and −6.6 % and does not improve with refinement; the finest of the nine rungs is the worst. Section 2.6 accounts for this, which is not a closure defect.

Table 9: Deviation from the exact arm (%) at 20 dpa, $I = V = 1000$, with $i_{\mathrm{discrete}} = v_{\mathrm{discrete}} = 50$ throughout. Only the bin count varies.

Rung	I_{bin}	r	N_{eq}	N_{111}	d_{111}	N_{100}	d_{100}	d_{cav}
B2	2	4.47	114	+3.7	-4.4	+1.2	-0.6	-2.7
B3	3	2.71	118	+1.8	-5.4	-2.1	+1.0	-2.1
B4	4	2.11	122	+1.4	-5.3	-1.2	-5.4	-2.4
B6	6	1.65	130	+1.2	-5.8	-2.1	-2.3	-1.6
B8	8	1.45	138	+1.1	-6.3	-1.2	-0.8	-0.8
B12	13	1.28	158	+1.0	-4.8	-0.2	+0.1	-0.1
B16	17	1.21	174	+0.9	-5.4	-0.1	+0.3	+0.1
B24	25	1.13	206	+1.0	-6.4	-0.2	+0.1	+0.1
B32	33	1.10	238	+1.0	-6.6	-0.2	+0.1	+0.1

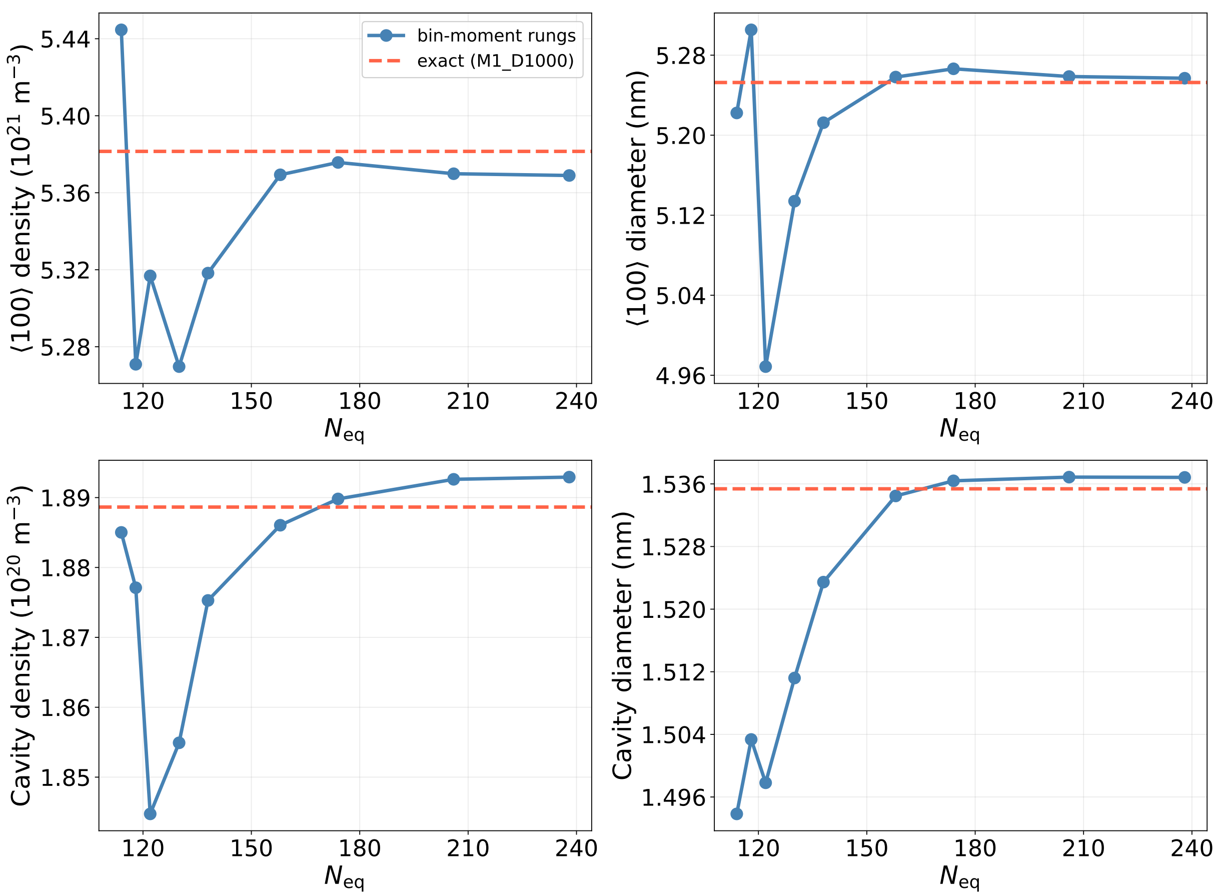

Figure 6: The $\langle 100 \rangle$ and cavity observables against N_{eq} at fixed $i_{\mathrm{discrete}} = 50$, the exact arm dashed. Each closes on the reference by $I_{\mathrm{bin}} = 12$, at $N_{\mathrm{eq}} = 158$. The $\frac{1}{2}\langle 111 \rangle$ panels are omitted; that population is given quantitatively in the table above.

2.5 The two-dimensional sweep

Tables 10 and 11 give the sweep in which both parameters vary independently.

Table 10: Deviation of d_{111} from the exact arm (%) at 20 dpa. Read down a column, not across a row: the error falls with i_{discrete} and is nearly flat in I_{bin}.

i_{discrete}	$I_{\text{bin}}=4$	$I_{\text{bin}}=8$	$I_{\text{bin}}=16$	$I_{\text{bin}}=32$
25	-7.2	*d.n.r.*	-6.7	-6.3
50	-5.3	-6.3	-5.4	-6.6
100	*d.n.r.*	-3.4	-4.2	-5.1
200	-6.1	-2.0	-2.5	-3.4
400	*d.n.r.*	-0.1	*d.n.r.*	-1.1
800	+1.0	+0.7	+0.6	*n/r*

Table 11: Deviation of d_{cav} (%) over the same sweep, shown for contrast: converged almost everywhere, with a single anomaly discussed below.

i_{discrete}	$I_{\text{bin}}=4$	$I_{\text{bin}}=8$	$I_{\text{bin}}=16$	$I_{\text{bin}}=32$
25	-2.7	*d.n.r.*	+0.1	+0.0
50	-2.4	-0.8	+0.1	+0.1
100	*d.n.r.*	-0.3	-0.0	+0.0
200	+15.1	-0.2	-0.1	-0.1
400	*d.n.r.*	-0.2	*d.n.r.*	-0.1
800	-0.1	-0.1	-0.0	*n/r*

Two entries require explanation. *d.n.r.* denotes a cell that reached the one-hour limit short of 20 dpa, a cost result rather than a failure of the method. *n/r* denotes a configuration the bin layout cannot realise: at $i_{\text{discrete}}=800$ only 200 sizes remain for the bins, so $I_{\text{bin}}=32$ would return 24, and the layout check raises rather than silently running a different grid — a guard added after nine bit-identical "refinement" runs were produced by a dropped bin configuration.

Down the column $I_{\text{bin}}=8$ the deviations are -6.3, -3.4, -2.0, -0.1 and $+0.7$ percent; across the row $i_{\text{discrete}}=100$ they are -3.4, -4.2 and -5.1, worsening with every bin added. The bottom row, $i_{\text{discrete}}=800$, is converged to better than 1 % in d_{111} at every bin count. The axis governing this error is therefore not the one the M2R ladder swept.

One cell is anomalous. At $i_{\text{discrete}}=200$, $I_{\text{bin}}=4$ the cavity diameter deviates by 15.1 % where its row-neighbours sit at -0.2 %, with N_{cav} and $\overline{n}_v$ both 52 %. It ran in 6 s against 39 min to 46 min for those neighbours and reproduces on re-run. Coarse bins render the system non-stiff, which accounts for the speed but not the error, since the coarser $I_{\text{bin}}=4$ cells at $i_{\text{discrete}}=25$ and 50 deviate by only -2.7 and -2.4 %. The cause is not established; it is recorded rather than smoothed over, and is a sharper instance of the non-monotonicity seen in Section 1.5.

2.6 Origin of the residual mean-size error

The M2R ladder varied the parameter that does not control this error: the $\frac{1}{2}\langle 111\rangle$ mean-size error converges in i_{discrete}, not I_{bin}, and $i_{\text{discrete}}=50$ was held fixed across all nine rungs. At $I_{\text{bin}}=16$

and 0.05 dpa, against an exact $d_{111} = 2.9787$ nm, the error tracks the fraction of $\frac{1}{2}\langle 111\rangle$ content lying above the discrete core — the fraction the closure must represent:

i_{discrete}		d_{111} (nm)	% of content binned
25	2.7232	(−8.6%)	96.6
50	2.7744	(−6.9%)	91.2
100	2.8128	(−5.6%)	83.4
200	2.8692	(−3.7%)	70.6
400	2.9403	(−1.3%)	47.7
800	2.9813	(+0.1%)	8.0

This is ordinary closure error, converging along the axis the M2R ladder held fixed. Conversely, at $i_{\text{discrete}} = 400$, raising I_{bin} from 4 to 8 to 33 gives 2.9615, 2.9503 and 2.9245 nm — slightly worse each time.

Nothing is special about the $\frac{1}{2}\langle 111\rangle$ population here: its mean size is about 147 interstitials, some three times the discrete core, so the peak of its distribution lies in the binned region and is carried by the closure. d_{100} and d_{cav} improve monotonically along the same sweep with about a tenth the amplitude.

Four alternative explanations were rejected by measurement. A difference between code paths is excluded by running the bin-moment path with $I_{\text{bin}} = 0$ and $i_{\text{discrete}} = I$, carrying no closure: it reproduces the discrete arm at 20 dpa to within rounding, d_{111} to −0.000%. `LOOP_COAL`, the leading candidate as the only channel implemented twice, is excluded because with it disabled in both arms the deficit is unchanged, −7.3% and −7.0% against −7.0% and −7.9% with it enabled. A domain-edge artefact is excluded because the exact spectrum decays smoothly to the top of the grid, size 1000 holding 0.01 % of the content. Slow accumulation is excluded because the deficit already stands at −8.4% at 0.005 dpa and drifts only to −6.6% by 20 dpa.

References

[1] C. Dethloff *et al.* (2016), “Microstructural defects in EUROFER 97 after different neutron irradiation conditions.” Local copy: `docs/Literature`.

[2] C. Dethloff *et al.* (2018), “Review and critical assessment of dislocation loop analyses on EUROFER 97,” *Nuclear Materials and Energy*, `doi:10.1016/j.nme.2018.05.015`.

[3] M. Klimenkov *et al.* (2011), “Characterization of radiation induced defects in EUROFER 97 after neutron irradiation,” *Journal of Nuclear Materials*, `doi:10.1016/j.jnucmat.2010.12.261`.

[4] M. Klimenkov *et al.* (2020), “Correlation of microstructural and mechanical properties of neutron irradiated EUROFER97 steel,” *Journal of Nuclear Materials*, `doi:10.1016/j.jnucmat.2020.152231`.

[5] O. J. Weiß *et al.* (2012), “Quantitative characterization of microstructural defects in up to 32 dpa neutron irradiated EUROFER97,” *Journal of Nuclear Materials*, `doi:10.1016/j.jnucmat.2012.03.027`.

[6] A. Chauhan *et al.* (2021), “Post-irradiation annealing of neutron-irradiated EUROFER97,” *Journal of Nuclear Materials*, `doi:10.1016/j.jnucmat.2021.152863`.

[7] R. Coppola *et al.* (2019), "Dose dependence of micro-voids distributions in low-temperature neutron irradiated Eurofer97 steel." Local copy: `docs/Literature`.